GetMaster — To'liq Texnik Topshiriq v2.0

Mukammallashtirilgan versiya — v1.0 asosida, barcha bo'shliqlar aniq qiymatlar bilan to'ldirilgan

2026-09-22 · @Someone

1. Kirish, maqsad va to'liq oqim

GetMaster — mijozning uy/kundalik xizmat muammosini AI yordamida tasniflab, mos ustalar bilan bog'laydigan va butun buyurtma umrini (booking'dan to'lov va kafolatgacha) boshqaradigan marketplace.

```
flowchart LR
  A[Muammo] --> B[So'rov]
  B --> C[AI tasnif]
  C --> D[Matching]
  D --> E[Takliflar]
  E --> F[Usta tanlash]
  F --> G[Booking]
  G --> H[Order]
  H --> I[Ish + Proof]
  I --> J[Tasdiq]
  J --> K[To'lov]
  K --> L[Kafolat]
  L --> M[Review]
```

Parallel quyi tizimlar har bir bosqichda ishlaydi: Chat, Notification, Trust & Safety, Dispute/Appeal, Audit. Ushbu v2.0 hujjat v1.0 TZ'ning barcha talablarini saqlagan holda, aniq raqamlar, kontraktlar va formulalar bilan to'ldiradi — maqsad: ishlab chiquvchi jamoa hech qanday talqin qilish zaruratisiz to'g'ridan-to'g'ri kodlay olishi.

2. Domen zonalari, rollar va ruxsatlar matritsasi

Zonalar: `/` (rol tanlash) · `/mijozlar/*` · `/ustalar/*` · `/admin/*` (himoyalangan). Har bir zona faqat o'z rolining route'larini render qiladi; noto'g'ri rol bilan kirishda 403 + role tanlash sahifasiga redirect.

Route map (v1.0'dan o'zgarishsiz): Mijoz — `/register, /login, /verify, /profil, /sorov-yaratish, /sorovlar, /sorovlar/[id], /takliflar, /buyurtmalar, /buyurtmalar/[id], /chat, /tolovlar, /kafolat, /shikoyatlar`. Usta — `/register, /login, /verify, /profil, /xizmatlar, /portfolio, /jadval, /verification, /sorovlar, /sorovlar/[id], /takliflar, /buyurtmalar, /buyurtmalar/[id],`

`/chat, /tolovlar, /statistika`. Admin — `/users, /masters, /requests, /orders, /offers, /payments, /disputes, /violations, /appeals, /services, /reviews, /legal, /audit, /notifications, /settings`.

Ruxsatlar matritsasi (asosiy amallar; — = taqiqlangan, R = faqat oʻziniki, R-all = barcha yozuvlar):

Amal	CUSTOMER	MASTER	ADMIN
Soʻrov yaratish/tahrirlash	R	—	R-all
Offer yuborish	—	R	—
Offer qabul qilish	R (oʻz soʻrovi)	—	—
Booking/Order koʻrish	R	R	R-all
Order status oʻzgartirish	faqat CONFIRM/DISPUTE	faqat ON_THE_WAY→COMPLETED oraligʻi	har qanday (audit bilan)
Work proof yuklash	—	R	—
Additional work soʻrash	—	R	—
Additional work tasdiqlash	R (oʻz orderi)	—	—
Toʻlov boshlash	R	—	—
Refund berish	—	—	R-all
Review yozish	R (completed order)	—	moderation (flag/hide)
Master verification koʻrib chiqish	—	—	R-all
Case (complaint/dispute) ochish	R	R	R-all
Case hal qilish/jazo qoʻllash	—	—	R-all
Policy versiyalarini yaratish	—	—	R-all

Amal	CUSTOMER	MASTER	ADMIN
Audit log ko'rish	—	—	R-all

Backend darajasida bu jadval `RolesGuard` + har bir service metodida **object-level tekshiruv** (masalan `order.customerId === currentUser.sub`) orqali ikki qatlamda amalga oshiriladi — faqat rol tekshiruvi yetarli emas.

3. Auth, sessiya va xavfsizlik — aniq parametrlar

Parametr	Qiymat
OTP kod uzunligi	6 raqam
OTP amal qilish muddati	5 daqiqa
OTP qayta yuborish (resend) kutish vaqti	60 soniya
OTP so'rovlar limiti	5 ta / telefon / 1 soat oynasi
OTP tekshirish urinishlari	5 marta, undan keyin kod bloklanadi
Access JWT TTL	900 soniya (15 daqiqa)
Refresh token TTL	30 kun
Max faol qurilmalar soni	2 ta (default), `MAX_ACTIVE_DEVICES` orqali sozlanadi
Parol talablari	kamida 8 belgi, bcrypt cost-factor 12
Refresh cookie	`HttpOnly; Secure; SameSite=Lax; Path=/api/v1/auth`
Login rate-limit	5 urinish / 15 daqiqa / (IP + telefon) juftligi
Registratsiya rate-limit	3 ta / soat / IP
Akkaunt vaqtincha bloklash	10 muvaffaqiyatsiz login / 1 soat → 15 daqiqa lock
Refresh-token reuse detection	eskirgan/qayta ishlatilgan refresh token aniqlansa — shu `familyId` ga tegishli barcha tokenlar zudlik bilan bekor qilinadi va foydalanuvchiga xavfsizlik ogohlantirishi yuboriladi

Registratsiya oqimi: `register` → status `PENDING` → OTP (`purpose=REGISTER`) yuboriladi → `verify-registration` → talab qilingan policy versiyalari (Terms, Privacy, Platform Rules) qabul qilingan bo'lsa → status `ACTIVE` .

Login oqimi: `login` (telefon+parol) → agar shubhali qurilma/IP bo'lsa OTP (`purpose=LOGIN`) talab qilinadi → muvaffaqiyatli bo'lsa access JWT (response body) + refresh token (HttpOnly cookie) qaytariladi, `DeviceSession` yaratiladi/yangilanadi.

Logout: joriy qurilma — faqat shu `RefreshToken` va `DeviceSession` bekor qilinadi; barcha qurilmalar — foydalanuvchining barcha faol session/token'lari bekor qilinadi va audit yozuvi yaratiladi.

4. Ma'lumotlar bazasi arxitekturasi va ERD

Asosiy agregatlar orasidagi bog'lanish:

```
erDiagram
  User ||--o| MasterProfile : "1:1 (MASTER role)"
  User ||--o{ ProblemRequest : yaratadi
  ProblemRequest ||--o{ Offer : oladi
  Offer ||--o| Booking : "accept => 1"
  Booking ||--|| Order : yaratadi
  Order ||--o{ OrderStatusHistory : logAudit
  Order ||--o{ WorkProof : proof
  Order ||--o{ Payment : to'lov
  Order ||--o| Warranty : kafolat
  Order ||--o| Review : baho
  Order ||--o{ Case : complaint_or_dispute
  MasterProfile ||--o{ MasterService : xizmatlar
  MasterProfile ||--o{ MasterVerification : hujjatlar
```

Domenlar bo'yicha model guruhlari (to'liq maydonlar — loyihaning `prisma/schema.prisma` fayli, 44 model, kanonik manba hisoblanadi):

Domen	Modellar
Identity	User, UserRole, RefreshToken, OtpCode, DeviceSession, PolicyVersion, PolicyAcceptance
Katalog/Usta	ServiceCategory, Service, MasterProfile, MasterService, MasterPortfolio, MasterAvailability, MasterVerification

Domen	Modellar
So'rov/Matching	ProblemRequest, ProblemAttachment, AIClassification, MatchingRun, MatchingResult, Offer, Booking
Order	Order, OrderStatusHistory, WorkProof, AdditionalWorkRequest
Aloqa	Conversation, ConversationParticipant, Message
Moliya	Payment, Refund, Warranty, WarrantyClaim
Trust & Safety	Review, Case, Evidence, UserAction, Appeal, Compensation
Operatsiyalar	Notification, AuditLog, IdempotencyKey, OutboxEvent

v1.0 TZ'dan qasddan chetlashishlar (loyihada allaqachon shunday qaror qilingan, rasmiylashtirilmoqda):

- TZ'dagi alohida `Violation` / `Penalty` o'rniga **`Case`** (COMPLAINT yoki DISPUTE turi bilan) + **`UserAction`** (WARNING/FINE/RESTRICTION/SUSPENSION/BAN) ishlatiladi — kontseptual ekvivalent, lekin kamroq jadval bilan.
- `MessageAttachment` alohida jadval emas, `Message.attachments` (Json) maydonida saqlanadi.
- Har bir `Decimal` moliyaviy maydon `@db.Decimal(14,2)` aniqligida (UZS uchun yetarli).

Muhim cheklov: `RefreshToken` da `@@unique([userId, deviceId])` qasddan olib tashlangan — rotation paytida eski (revoked) va yangi token bir vaqtda saqlanishi kerak (reuse detection uchun).

5. AI Classification — integratsiya kontrakti

AI — yakuniy hakam emas, faqat tavsiya beradi. So'rov yuborilgach BullMQ `ai-classification` navbatiga vazifa qo'yiladi (async, sinxron API javobini bloklamaydi).

So'rov (worker → AI provider):

```
{
  "requestId": "uuid",
  "title": "string",
  "description": "string",
  "attachmentUrls": ["signed-url"],
  "locale": "uz"
}
```

Javob (AI provider → worker):

```
{
  "categoryId": "uuid|null",
  "serviceId": "uuid|null",
  "confidence": 0.0,
  "possibleQuestions": ["string"],
  "alternatives": [{"serviceId": "uuid", "confidence": 0.0}]
}
```

Qoidalar:

Holat	Amal
`confidence >= 0.75`	Kategoriya/xizmat avtomatik biriktiriladi, mijozga "AI aniqladi" belgisi bilan ko'rsatiladi
`0.4 <= confidence < 0.75`	Tavsiya sifatida ko'rsatiladi, mijoz tasdiqlashi yoki o'zgartirishi kerak
`confidence < 0.4` yoki AI javob bermadi	Mijozdan qo'lda kategoriya tanlash so'raladi (`possibleQuestions` fallback ro'yxati bilan)
AI timeout (>10s) yoki xato	2 marta qayta urinish (exponential backoff: 2s, 8s), muvaffaqiyatsiz bo'lsa — qo'lda tanlash rejimiga o'tadi, so'rov bloklanmaydi

Har bir AI javobi `AIClassification` jadvaliga yoziladi (`isAccepted`, `correctedAt` orqali mijoz/tizim tuzatishi kuzatiladi — kelajakda model sifatini o'lchash uchun).

6. Matching Engine — ball hisoblash formulasi

Hard filters (bittasi o'tmasa usta nomzodlikdan chiqadi): `MasterService.isActive=true` va `serviceId` mos ⑧ masofa ≤ `MasterProfile.workRadiusKm` ⑧ `MasterAvailability` bo'yicha vaqt oynasida band emas ⑧ `MasterProfile.status=ACTIVE` va `isPaused=false` ⑧ `verification=VERIFIED`.

Hard filterdan o'tgan nomzodlar uchun ball:

```
Score = w1*R + w2*(1 - d/dmax) + w3*A + w4*(rating/5) + w5*C + w6*P - w7*K
```

Belgi	Ma'no	Og'irlik
R	service relevance (0–1), AI confidence yoki exact-match	w1 = 0.25
d/dmax	masofa nisbati: distanceKm / workRadiusKm	w2 = 0.20
A	availability (0 yoki 1), joriy vaqt oynasida bo'shmi	w3 = 0.10
rating/5	ratingAvg / 5	w4 = 0.20
C	completionRate (0–1)	w5 = 0.15
P	responseRate (0–1)	w6 = 0.10
K	recent cancellation penalty (0–1), oxirgi 30 kunda bekor qilishlar nisbati	w7 = 0.15

Og'irliklar yig'indisi w1..w6 = 1.0; w7 jarima sifatida qo'shimcha ayiriladi. Algoritm versiyasi (`MatchingRun.algorithmVersion`, masalan "v1.0") har bir ishga tushirishda yoziladi va audit qilinadi — og'irliklar o'zgarsa versiya oshiriladi.

Taklif to'lqinlari (notification waves): eng yuqori balldan top-5 ustaga bir vaqtda push yuboriladi → 5 daqiqa ichida javob (offer) kelmasa, navbatdagi top-5 qo'shiladi → maksimal 3 to'lqin (jami 15 ta) yoki `ProblemRequest.expiresAt` gacha (default: yaratilgandan 24 soat keyin, `status=EXPIRED`).

7. Offer, Booking va concurrency (race-condition) strategiyasi

Offer: `price, estimatedMinutes, message, availableAt, expiresAt` (default TTL 2 soat). Status: `PENDING → ACCEPTED | REJECTED | EXPIRED | WITHDRAWN`.

Race condition muammosi: bitta `ProblemRequest` ga bir nechta offer kelishi mumkin; mijoz bittasini accept qilganda, boshqalari avtomatik bekor qilinishi kerak, va ikki parallel so'rov (masalan ikki marta bosish) bitta offerni ikki marta accept qilmasligi kerak.

Yechim — bitta Prisma `$transaction` ichida shartli yangilash:

```
BEGIN;
UPDATE offers SET status='ACCEPTED'
  WHERE id = :offerId AND status='PENDING' AND expires_at > now();
-- affected rows tekshiriladi: 0 bo'lsa => 409 Conflict "Offer allaqachon hal
qilingan"
UPDATE offers SET status='REJECTED'
  WHERE request_id = :requestId AND id != :offerId AND status='PENDING';
UPDATE problem_requests SET status='BOOKED' WHERE id = :requestId;
INSERT INTO bookings (...) VALUES (...);
COMMIT;
```

Bu — optimistic conditional update (`WHERE status='PENDING'`) orqali PostgreSQL row-level lock avtomatik ta'minlanadi; qo'shimcha `SELECT ... FOR UPDATE` shart emas.

Idempotency: `POST /offers/:id/accept` va boshqa barcha critical mutation'lar `Idempotency-Key` header qabul qiladi — `IdempotencyKey` jadvalida `(key, endpoint)` unique; xuddi shu key bilan qayta so'rov kelsa, saqlangan `responseStatus` / `responseBody` qaytariladi (operatsiya qayta bajarilmaydi). Kalit TTL: 24 soat.

Booking: `Offer` accept qilingan payt `locationSnapshot` va `serviceSnapshot` (JSON) saqlanadi — keyinchalik manzil/narx o'zgarsa ham buyurtma tarixi buzilmaydi.

8. Order lifecycle — status o'tish matritsasi

```
stateDiagram-v2
  [*] --> BOOKED
  BOOKED --> CONFIRMED
  CONFIRMED --> ON_THE_WAY
  ON_THE_WAY --> ARRIVED
  ARRIVED --> IN_PROGRESS
  IN_PROGRESS --> WAITING_CONFIRMATION
  WAITING_CONFIRMATION --> COMPLETED
  BOOKED --> CANCELLED
  CONFIRMED --> CANCELLED
  BOOKED --> EXPIRED
  IN_PROGRESS --> DISPUTED
  WAITING_CONFIRMATION --> DISPUTED
  DISPUTED --> COMPLETED
  DISPUTED --> CANCELLED
```

Eski status	Yangi status	Ruxsat etilgan actor	Shart
BOOKED	CONFIRMED	MASTER	booking qabul qilingach 30 daqiqa ichida
BOOKED	EXPIRED	SYSTEM	master 30 daqiqada tasdiqlamasa
CONFIRMED	ON_THE_WAY	MASTER	
ON_THE_WAY	ARRIVED	MASTER	geolokatsiya yoki qo'lda belgilash
ARRIVED	IN_PROGRESS	MASTER	kamida 1 ta BEFORE work-proof yuklangach
IN_PROGRESS	WAITING_CONFIRMATION	MASTER	kamida 1 ta AFTER work-proof yuklangach
WAITING_CONFIRMATION	COMPLETED	CUSTOMER	mijoz tasdiqlaydi (yoki 48 soatdan keyin SYSTEM avtomatik tasdiqlaydi)
WAITING_CONFIRMATION / IN_PROGRESS	DISPUTED	CUSTOMER, MASTER	Case ochiladi
BOOKED / CONFIRMED	CANCELLED	CUSTOMER, MASTER, ADMIN	sabab majburiy (`cancellationReason`)
DISPUTED	COMPLETED / CANCELLED	ADMIN	Case yopilgach

Har bir o'tish `OrderStatusHistory` ga `actorId, actorType, oldStatus, newStatus, reason` bilan yoziladi. Server jadvalda ko'rsatilmagan o'tishni **qat'iy rad etadi** (409, `INVALID_TRANSITION`) — bu holat mashinasi service qatlamida markazlashtirilgan (`OrderStateMachine`) bo'lishi kerak, controller'da emas.

9. Work Management — Additional Work va bekor qilish qoidalari

WorkProof: har bir order uchun kamida 1 ta `BEFORE` (ish boshlanishidan oldin) va 1 ta `AFTER` (tugagach) proof majburiy; `DURING` ixtiyoriy. Fayl: foto/video, min 1 ta max 10 ta har bosqich uchun.

AdditionalWorkRequest biznes qoidalari:

Qoida	Qiymat
Kim so'raydi	faqat MASTER, faqat order `IN_PROGRESS` holatida

Qoida	Qiymat
Majburiy maydonlar	`reason, description, amount`, kamida 1 ta evidence (foto)
Max qo'shimcha summa	asl `agreedPrice` ning 50%igacha tasdiqsiz taklif qilinishi mumkin; undan yuqori bo'lsa qo'shimcha ADMIN xabardor qilinadi (fraud monitoring)
Mijoz javob muddati	2 soat (`expiresAt`); javob bo'lmasa `status=EXPIRED`, master ishni davom ettira olmaydi to'liq tasdiq bo'lmaguncha
Narx oshirilishi	**faqat** `status=APPROVED` bo'lgach `Order.finalPrice` ga qo'shiladi; tasdiqsiz hech qanday narx o'zgarishi bo'lmaydi

Bekor qilish (cancellation) qoidalari (TZ v1.0'da raqamlanmagan, shu yerda belgilanadi):

Bosqich	Mijoz tomonidan bekor qilish	Jarima
BOOKED (master hali CONFIRM qilmagan)	bepul	yo'q
CONFIRMED, lekin `scheduledStartAt` gacha > 2 soat	bepul	yo'q
CONFIRMED, `scheduledStartAt` gacha ⩽ 2 soat	mumkin	usta uchun kompensatsiya — `agreedPrice` ning 10% (Compensation yozuvi orqali)
ON_THE_WAY / ARRIVED dan keyin	faqat dispute orqali	ADMIN qaror qiladi

Master tomonidan asossiz bekor qilish `cancellationRate` ga ta'sir qiladi va 30 kunda 3 martadan ortiq bo'lsa avtomatik `RESTRICTED` statusiga o'tkaziladi (Trust & Safety, §12 bilan bog'liq).

10. Chat, media va storage arxitekturasi

Storage: S3-compatible — production'da MinIO (self-hosted) yoki AWS S3. Bucket tuzilishi: `getmaster-media/{domain}/{entityId}/{uuid}.{ext}` (masalan `getmaster-media/work-proofs/{orderId}/{uuid}.jpg`).

Media turi	Max hajm	Ruxsat etilgan format
Photo	10 MB	jpg, jpeg, png, webp, heic
Video	100 MB	mp4, mov
Voice	10 MB	ogg, m4a, mp3
Document	20 MB	pdf, doc, docx

Tekshirish zanjiri (yuklashda): kengaytma → MIME-type (haqiqiy fayl signature bo'yicha, faqat extension emas) → hajm → (rasm bo'lsa) Sharp orqali qayta ishlash: `thumbnail 200×200`, `medium 800×800`, original saqlanadi, EXIF metadata olib tashlanadi (privacy).

Object-level authorization: har bir media so'rovida backend tekshiradi — so'rovchi shu order/conversation/request ishtirokchisimi (customer/master) yoki ADMIN'mi. Public S3 URL hech qachon qaytarilmaydi.

Signed URL: GET uchun 15 daqiqa TTL, faqat o'qish huquqi bilan (presigned GET). Har bir chaqiriqda yangi URL generatsiya qilinadi, keshlanmaydi.

Conversation turlari: `ORDER` (order bilan 1:1 bog'liq, `orderId` unique) va `SUPPORT` (mijoz/usta ↔ ADMIN). Xabar tezligi cheklovi: 30 xabar/daqiqa/foydalanuvchi (spam himoyasi).

11. Payment, Refund va Warranty

Provayderlar (`PaymentProvider` enum): **Click**, **Payme**, **Uzum** (O'zbekiston bozori uchun standart uchlik), + `OTHER` kengaytirish uchun.

Oqim: `POST /payments` → `Payment(status=PENDING)` yaratiladi → provayderning checkout/invoice URL'iga redirect → provayder webhook orqali natija yuboradi (`POST /webhooks/payments/:provider`) → webhook imzosi (`provider` ga xos HMAC/checksum) tekshiriladi → `status=SUCCEEDED|FAILED` + `providerPaymentId` yoziladi → `OutboxEvent` orqali `order.payment.succeeded` eventi chiqariladi.

Idempotency: har bir `Payment` yaratish so'rovi `Idempotency-Key` bilan himoyalangan; webhook'lar tabiatan idempotent qilinadi — `(provider, providerPaymentId)` unique constraint orqali takror webhook e'tiborsiz qoldiriladi.

Refund: `Refund.idempotencyKey` unique. To'liq yoki qisman (`amount < Payment.amount`); qisman refund'lar yig'indisi `Payment.refundedAmount` ni oshira olmaydi (service qatlamida tekshiriladi). Kutilayotgan bank muddati: 3–10 ish kuni (foydalanuvchiga ko'rsatiladi, lekin `Refund.status` shu oraliqda `PROCESSING` da qoladi).

Warranty: `MasterService` da xizmat turiga qarab sozlanadigan muddat, default **30 kun** (`Warranty.startsAt = Order.completedAt`, `expiresAt = startsAt + 30 kun`). `WarrantyClaim` ochish oynasi — faqat kafolat muddati ichida, `OPEN → UNDER_REVIEW → APPROVED/REJECTED → RESOLVED`. APPROVED bo'lsa — ADMIN qaroriga ko'ra `Compensation` (REFUND/PARTIAL_REFUND/CREDIT) yaratiladi.

Order va Payment statuslari mustaqil: masalan `Order.status=DISPUTED` bo'lishi mumkin, `Payment.status=SUCCEEDED` bo'lgan holda ham — ular hech qachon bitta enum'ga birlashtirilmaydi.

12. Review, Reputation va Trust & Safety

Review: faqat `Order.status=COMPLETED` bo'lgach, `authorId` shu order ishtirokchisi bo'lsagina yaratiladi (`@@unique([orderId])` — order boshiga bitta review). `rating` 1–5 butun son.

Reputation qayta hisoblash (Review yaratilgan/o'zgargan har safar, background job orqali, synxron emas):

```
ratingAvg_new = SUM(rating) / COUNT(rating)  over last 200 PUBLISHED reviews
```

So'nggi 200 review bilan cheklash — eski xatti-harakat uzoq muddat reputatsiyaga cheksiz ta'sir qilmasligi uchun. `ratingCount` — umumiy PUBLISHED review soni (cheklanmagan).

Shubhali review flag qilinishi: avtomatik trigger — rating=1 va sharh matni yo'q, yoki bitta IP'dan 24 soatda 5+ review — `status=FLAGGED`, ADMIN moderatsiyasigacha ommaviy ko'rinishda pasaytirilgan vaznda hisoblanadi.

Trust & Safety jazo matritsasi (`UserAction.type` bo'yicha, `Case` `CONFIRMED` bo'lgach ADMIN tomonidan qo'llanadi):

Buzilish darajasi	1-marta	2-marta (180 kun ichida)	3-marta va undan ko'p
Yengil (kechikish, javobsizlik)	WARNING	RESTRICTION (7 kun, `OFFERS`)	SUSPENSION (14 kun)
O'rtacha (bekor qilish, past sifat)	RESTRICTION (7 kun)	SUSPENSION (14 kun)	SUSPENSION (30 kun)
Og'ir (firibgarlik, xavfsizlik buzilishi, haqorat)	SUSPENSION (30 kun)	BAN	BAN

Har bir `UserAction` `PolicyVersion` ga bog'lanadi (qaysi qoida asosida jazolangani aniq bo'lishi uchun) va foydalanuvchi 14 kun ichida `Appeal` bera oladi — `Appeal.status: OPEN → UNDER_REVIEW → ACCEPTED|REJECTED → CLOSED`.

13. Notifications — kanal marshrutlash

Barcha notification'lar BullMQ `notifications` navbatiga yoziladi (`Notification.status: PENDING→SENT|FAILED`), request javobini bloklamaydi.

Event	Default kanallar	Fallback
OTP kod	SMS	— (kritik, muqobil yo'q)
Yangi so'rov (usta uchun)	PUSH + IN_APP	PUSH 2 daqiqada yetmasa → SMS
Offer kelishi (mijoz uchun)	PUSH + IN_APP	—
Offer qabul qilindi (usta uchun)	PUSH + IN_APP + TELEGRAM (agar ulangan bo'lsa)	—
Booking/Order status o'zgarishi	IN_APP + PUSH	—
Usta yo'lda / yetib keldi	PUSH (yuqori prioritet)	—
Additional work so'rovi	PUSH + SMS	mijoz 1 soat javob bermasa qayta PUSH
Completion / to'lov muvaffaqiyatli	IN_APP + EMAIL (agar email bo'lsa)	—
Warranty muddati tugashi (7 kun oldin)	IN_APP + PUSH	—
Dispute/Appeal holati o'zgarishi	IN_APP + EMAIL	—
Kritik xavfsizlik (yangi qurilmadan login, reuse detection)	SMS + EMAIL	— (darhol, navbatdan tashqari)

Qayta urinish siyosati: 3 marta, backoff 1/5/15 daqiqa; barchasi muvaffaqiyatsiz bo'lsa `status=FAILED` + `error` maydoniga sabab yoziladi, monitoring metrikaga chiqadi (§16).

14. To'liq API spetsifikatsiyasi

Base URL: `/api/v1`. Barcha javoblar bitta envelope'da:

```
{ "success": true, "data": { }, "requestId": "uuid" }
```

```
{ "success": false, "error": { "statusCode": 400, "code": "VALIDATION_ERROR",
"message": "string", "requestId": "uuid" } }
```

Standart xato kodlari: `VALIDATION_ERROR` (400) · `UNAUTHORIZED` (401) · `FORBIDDEN` (403) · `NOT_FOUND` (404) · `CONFLICT` (409, masalan offer allaqachon hal qilingan) · `INVALID_TRANSITION` (409, order status) · `RATE_LIMITED` (429) · `INTERNAL_ERROR` (500).

Pagination: cursor-based, `?cursor=<id>&limit=20` (default 20, max 100), javobda `{ items: [], nextCursor: string|null }`.

Metod + yo'l	Auth	Rol	Tavsif
POST /auth/register, /login, /verify-otp, /verify-registration, /refresh	Public	—	§3
POST /auth/logout, GET /auth/sessions, POST /auth/sessions/:deviceId/revoke	JWT	any	§3
GET/PATCH /users/me	JWT	any	profil
POST /addresses, GET/PATCH/DELETE /addresses/:id	JWT	CUSTOMER	manzillar
GET /services, /service-categories	Public	—	katalog
POST/GET/PATCH /master/profile, /master/services, /master/portfolio, /master/availability	JWT	MASTER	§7
POST /master/verification, GET /master/verification	JWT	MASTER	hujjat yuklash
PATCH /admin/master-verifications/:id	JWT	ADMIN	approve/reject/correction
POST/GET /problem-requests, GET /problem-requests/:id, PATCH /problem-requests/:id, POST /problem-requests/:id/cancel	JWT	CUSTOMER	§9
GET /problem-requests/:id/classification	JWT	CUSTOMER	AI natijasi, §5
PATCH /problem-requests/:id/classification	JWT	CUSTOMER	qo'lda tuzatish
GET /problem-requests/:id/offers, POST /problem-requests/:id/offers (master yaratadi)	JWT	CUSTOMER (GET), MASTER (POST)	§6–§7
POST /offers/:id/accept, POST /offers/:id/withdraw	JWT + Idempotency-Key	CUSTOMER / MASTER	§7

Metod + yo'l	Auth	Rol	Tavsif
GET /bookings/:id, GET /orders, GET /orders/:id	JWT	ishtirokchi	§8
POST /orders/:id/status	JWT	MASTER/CUSTOMER/ADMIN	§8 status matritsasiga qat'iy mos
POST /orders/:id/work-proofs	JWT	MASTER	§9
POST /orders/:id/additional-work, POST /additional-work/:id/approve|reject	JWT + Idempotency-Key	MASTER (create) / CUSTOMER (approve)	§9
POST /orders/:id/confirm	JWT	CUSTOMER	WAITING_CONFIRMATION→COMPLETED
POST /payments, GET /payments/:id, POST /webhooks/payments/:provider	JWT (webhook — provider signature)	CUSTOMER	§11
POST /payments/:id/refund	JWT	ADMIN	§11
GET /orders/:id/warranty, POST /warranty-claims, PATCH /warranty-claims/:id	JWT	CUSTOMER / ADMIN	§11
POST /orders/:id/review, GET /masters/:id/reviews	JWT	CUSTOMER	§12
POST /cases (complaint/dispute), GET /cases/:id, POST /cases/:id/evidence	JWT	CUSTOMER/MASTER	§12
PATCH /admin/cases/:id, POST /admin/cases/:id/actions	JWT	ADMIN	jazo qo'llash
POST /appeals, PATCH /admin/appeals/:id	JWT	any / ADMIN	§12
GET/POST conversations, GET/POST /conversations/:id/messages	JWT	ishtirokchi	§10
GET /notifications, PATCH /notifications/:id/read	JWT	any	§13
GET /admin/audit-logs, GET /admin/users, PATCH /admin/users/:id/status	JWT	ADMIN	§2
GET /health, GET /health/ready	Public	—	§16

Har bir mutation endpoint uchun majburiy zanjir: **AuthGuard → RolesGuard → object-level tekshiruv (service ichida) → DTO validation → (kerak bo'lsa) Idempotency-Key → biznes logikasi → AuditLog/OutboxEvent**.

15. Frontend arxitekturasi (Next.js)

```
app/
  (auth)/register/, login/, verify/
  (customer)/mijozlar/{profil,sorov-
yaratish,sorovlar,sorovlar/[id],takliflar,buyurtmalar,buyurtmalar/[id],chat,t
olovlar,kafolat,shikoyatlar}

(master)/ustalar/{profil,xizmatlar,portfolio,jadval,verification,sorovlar,sor
ovlar/[id],takliflar,buyurtmalar,buyurtmalar/[id],chat,tolovlar,statistika}

(admin)/admin/{users,masters,requests,orders,offers,payments,disputes,violati
ons,appeals,services,reviews,legal,audit,notifications,settings}
  api/ (route handlers — faqat BFF proxy, biznes logika yo'q)
middleware.ts        # role-aware routing, cookie'dagi rol/permission
tekshiruvi
lib/api-client.ts    # markazlashtirilgan fetch wrapper
lib/auth.ts           # access token faqat in-memory (Zustand); refresh token
HAR QACHON localStorage'da SAQLANMAYDI
```

Holat boshqaruvi: server state — React Query (cache, invalidation); client/UI state — Zustand. Access token faqat brauzer xotirasida (module-level variable), sahifa yangilanganda `/auth/refresh` orqali (HttpOnly cookie yordamida) tiklanadi.

API client xatti-harakati: 401 kelsa — bitta marta avtomatik `/auth/refresh` chaqiriladi, muvaffaqiyatli bo'lsa so'rov qayta yuboriladi; ikkinchi marta 401 — login sahifasiga redirect. Parallel so'rovlarda refresh faqat bitta marta ishga tushishi uchun in-flight promise ulashiladi.

Har bir sahifada majburiy holatlar (TZ v1.0 §24'dan): loading (skeleton), empty, validation error, unauthorized (401→redirect), not-found (404), conflict (409 — masalan offer band bo'lib qolgan), expired (masalan OTP/offer muddati), server-error (500, retry tugmasi bilan), offline (network banner).

PWA: `manifest.webmanifest` + service worker faqat statik asset cache uchun (API javoblari cache qilinmaydi — moliyaviy/holat ma'lumotlari doim fresh bo'lishi kerak).

16. Performance, Observability va rate-limit qiymatlari

Latency byudjeti (p95):

Amal turi	Maqsad
Oddiy CRUD (GET/POST, AI/tashqi chaqiruvsiz)	< 300 ms

Amal turi	Maqsad
Matching (DB-only, tashqi chaqiruvsiz)	< 1.5 s
AI classification (async, foydalanuvchi kutmaydi)	< 10 s (worker ichida)
To'lov webhook qayta ishlash	< 2 s

Rate-limit jadvali (Redis token-bucket, IP + userId bo'yicha):

Endpoint guruhi	Limit
/auth/login	5 / 15 daqiqa / (IP+telefon)
/auth/register	3 / soat / IP
/otp/*	5 / soat / telefon (§3)
Umumiy GET	120 / daqiqa / user
Umumiy mutation (POST/PATCH)	60 / daqiqa / user
/problem-requests (yaratish)	10 / kun / CUSTOMER
/offers (yaratish)	30 / kun / MASTER

DB va query: barcha filtrlash/sortlash DB darajasida (`WHERE` / `ORDER BY`, ilovada emas); N+1 oldini olish uchun Prisma `include` / `select` aniq belgilanadi, review-checklist'da tekshiriladi; cursor pagination barcha list endpoint'larda majburiy; ochiq (public) endpoint'lar (`/services`, master public profil) shaxsiy maydonlarni (`phone`, `email`) hech qachon qaytarmaydi — alohida `PublicMasterDto` orqali.

Redis cache: faqat o'qish ko'p, o'zgarish kam bo'lgan ma'lumotlar uchun (`service-categories`, `services` — TTL 10 daqiqa), yozishda aniq invalidatsiya (`del` key), "stale-while-revalidate" ishlatilmaydi moliyaviy/holat ma'lumotlarida.

Observability: `GET /health` (jarayon tirikligi) va `GET /health/ready` (DB+Redis ulanishi) — K8s/Docker healthcheck uchun. Har bir log yozuvi JSON, `requestId` bilan bog'langan. Metrikalar (Prometheus formatida `/metrics`): `http_request_duration_seconds`, `queue_depth{queue}`, `otp_abuse_total`, `order_status_transition_total{from,to}`, `payment_failure_total{provider}`. Kritik alert (payment failure rate > 5% / 5 daqiqa, queue depth > 1000) — Telegram bot orqali on-call kanalga.

17. Deployment, Infrastructure, CI/CD va Backup

```
flowchart LR
  U[Internet] --> N[Nginx / HTTPS]
  N --> FE[Next.js]
  N --> BE[NestJS API]
  BE --> PG[(PostgreSQL)]
  BE --> R[(Redis)]
  BE --> S3[(Object Storage)]
  BE --> SMS[SMS provider]
  BE --> AI[AI provider]
  BE --> PAY[Click/Payme/Uzum]
```

Muhitlar: `development` (lokal, Docker Compose) → `staging` (production'ga o'xshash, sinov ma'lumotlari) → `production`. Har biri alohida DB, Redis, secrets.

CI pipeline (har bir PR uchun): `lint → typecheck → unit tests → integration tests (Postgres+Redis service container) → build → (main branch) e2e tests`. Har bir bosqich muvaffaqiyatsiz bo'lsa merge bloklanadi.

CD pipeline: `staging` ga avtomatik deploy (main branch push) → smoke test → qo'lda tasdiqlash → `production` ga deploy (`prisma migrate deploy`, **hech qachon `db push` ishlatilmaydi**) → health-check kutish → rollback tayyor (oldingi image tag saqlanadi).

Backup siyosati:

Nima	Chastota	Saqlash muddati
PostgreSQL to'liq backup	kuniga 1 marta	30 kun
PostgreSQL WAL (point-in-time recovery)	uzluksiz	7 kun
Object storage (media)	haftada 1 marta snapshot	90 kun
Restore test	oyiga 1 marta, staging'da	—

Secrets: faqat environment o'zgaruvchilari yoki secret manager (masalan Vault/AWS Secrets Manager) orqali; `.env` yoki tokenlar hech qachon repository/ZIP/Docker image ichiga kiritilmaydi (`.dockerignore`, `.gitignore` orqali qat'iy nazorat).

18. Testing strategiyasi va Definition of Done

Test turi	Qamrov maqsadi	Muhit
Unit (domain/service)	⩾ 70% branch coverage, kritik modullarda (auth, payment, matching, order state machine) ⩾ 90%	in-memory, DB'siz
Integration	har bir mutation endpoint uchun happy-path + kamida 2 ta failure-path (validation, authorization, conflict)	real PostgreSQL + Redis (Docker service container)
E2E	§1'dagi to'liq oqim: register→OTP→rules→request→AI→matching→offer→accept→booking→order→work→payment→warranty→review	staging'ga yaqin muhit
Authorization	har rol × har endpoint kombinatsiyasi uchun "ruxsat berilmagan" holat tekshiriladi (§2 matritsasi asosida)	integration
Race-condition	2 ta parallel offer-accept so'rovi — faqat bittasi yutishi tekshiriladi (§7)	integration
Idempotency	bir xil `Idempotency-Key` bilan 2 marta so'rov — operatsiya bir marta bajarilgani tekshiriladi	integration
Payment critical-path	webhook signature noto'g'ri bo'lsa rad etilishi, qisman refund summasi oshib ketmasligi	integration
File security	ruxsatsiz MIME/kengaytma, hajm limiti, ownership tekshiruvi	integration
Regression	har release oldidan to'liq E2E suite	CI (main branch)

Definition of Done (har bir funksiya uchun barchasi bajarilishi shart):

- [ ] Frontend UI (loading/empty/error holatlar bilan)
- [ ] Backend controller + service + DTO validation
- [ ] Prisma migration (agar sxema o'zgargan bo'lsa)
- [ ] Authorization (rol + object-level) tekshiruvi
- [ ] Xato holatlari (`§14` standart formatda)
- [ ] Kerakli AuditLog / OutboxEvent / Notification
- [ ] Unit + integration testlar (yuqoridagi jadval bo'yicha)
- [ ] Structured logging va tegishli metrikalar

- [ ] Hujjatlashtirilgan (bu TZ yoki API reference yangilangan)

"Faqat UI tayyor" — tayyor emas deb hisoblanadi.

19. Legal, muhim prinsiplar va yakuniy arxitektura

Legal: Terms, Privacy, Public Offer, Platform Rules, Payment/Refund/Cancellation/Warranty qoidalari O'zbekiston qonunchiligiga muvofiq ishlab chiqiladi va launchdan oldin mahalliy yurist tomonidan ko'rib chiqiladi. Har bir qoida `PolicyVersion` orqali versiyalanadi; foydalanuvchining qaysi versiyani qachon qabul qilgani `PolicyAcceptance` da saqlanadi (§3).

Muhim prinsiplar (qat'iy, hech qanday modul buzmasligi kerak):

#	Prinsip
1	Customer va Master lifecycle'lari o'z `/mijozlar` va `/ustalar` zonalarida qoladi
2	AI yakuniy qaror emas — confidence < 0.75 da inson tasdig'i shart (§5)
3	Report avtomatik aybdorlik emas — `Case` har doim `REPORTED→UNDER_REVIEW` orqali o'tadi
4	Payment va Order statuslari mustaqil boshqariladi (§11)
5	Review source-of-truth, `ratingAvg` undan hosila hisoblanadi (§12)
6	Critical mutationlarda `Idempotency-Key` majburiy (§7, §11)
7	Private ma'lumot (`phone` , `email`) public profilga chiqmaydi (§16)
8	Production secrets repository/ZIPga kiritilmaydi (§17)
9	Production DB faqat migrations bilan boshqariladi, `db push` ishlatilmaydi (§17)
10	Backup oyiga bir marta restore orqali tekshiriladi (§17)
11	Har bir ADMIN o'zgarishi `AuditLog` ga yoziladi (§2)

#	Prinsip
12	Order status o'tishlari faqat serverdagi state machine orqali tekshiriladi, client'ga ishonilmaydi (§8)

Yakuniy arxitektura:

```
flowchart TD
  USER --> MP[Master Profile]
  MP --> SVC[Service]
  USER --> PR[Problem Request]
  PR --> MATCH[Matching]
  MATCH --> OFF[Offer]
  OFF --> BOOK[Booking]
  BOOK --> ORD[Order]
  ORD --> WP[Work Proof]
  WP --> AW[Additional Work]
  AW --> PAY[Payment]
  PAY --> WAR[Warranty]
  WAR --> REV[Review]
  REV --> REP[Reputation]
  ORD -.-> CHAT[Chat]
  ORD -.-> NOTIF[Notification]
  ORD -.-> TS[Trust and Safety]
  TS -.-> DISP[Dispute]
  DISP -.-> APP[Appeal]
  ORD -.-> AUDIT[Audit]
```

Arxitektura kelajakda mobil ilova, Telegram Mini App, yangi to'lov provayderlar, AI modellar va yangi xizmat kategoriyalarini qo'shishga tayyor — modular monolith tuzilishi (§21) buni microservice'larga erta bo'linmasdan ta'minlaydi.